

Food Delivery System: Database Design & Architecture

ER DIAGRAM AND RELATIONAL SCHEMA

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System Description

The system begins with the Customer, who manages multiple delivery addresses and selects a Restaurant to browse its unique menu. Once the Order is placed, the restaurant prepares the food and updates the Order Status in real-time.

When the food is ready, a Delivery Driver assigns the order to themselves, picks it up, and brings it to the customer's door. Finally, to ensure quality, the customer provides Ratings for both the meal and the delivery service, closing the loop of a food delivery experience.

System Requirements

- Place order: Customers can choose a restaurant and place an order for food.
- Menu items: Restaurants offer specific menus and prices. (Out of scope: customising ingredients).
- Delivery time: Customers can specify a delivery time, either ASAP or scheduled for later.
- Delivery address: Customers can store multiple delivery addresses on their profile.
- Payment method (Added Feature): Customers must select a payment method (e.g., card or cash) to securely complete their order.
- Order status: Restaurants can accept the order and update its status in real-time as it is prepared.
- Assign & Pick up: Delivery drivers can claim an order, pick it up from the restaurant, and deliver it.
- Ratings: Customers can rate both the delivery drivers and the restaurants.

Database Architecture: Core Concepts

Entity (Strong): Independent objects in our system.

- Examples in our schema: CUSTOMER, RESTAURANT, FOOD_ORDER

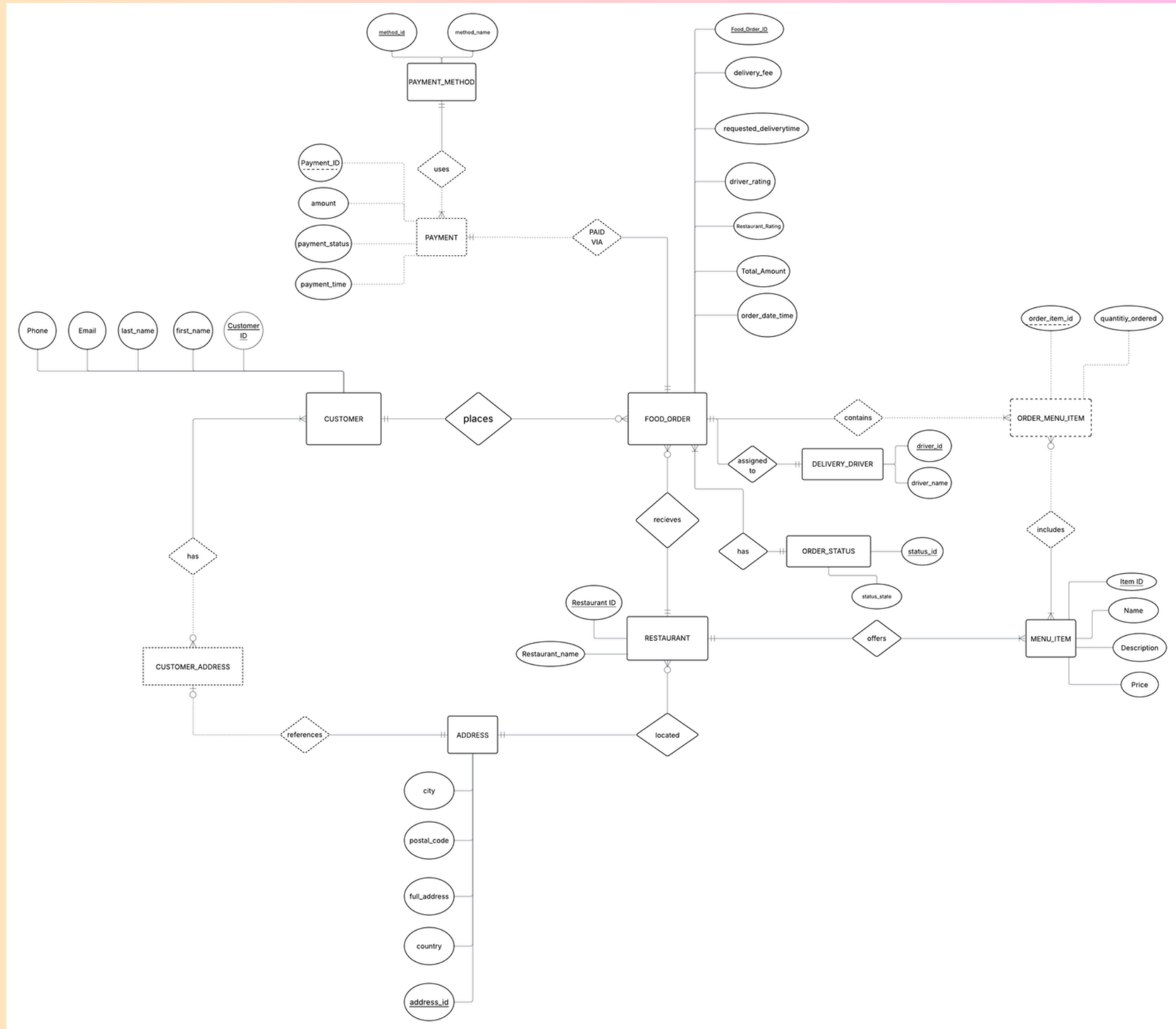
Attribute: Specific data fields describing an entity.

- Examples in our schema: first_name, Delivery_fee, Price

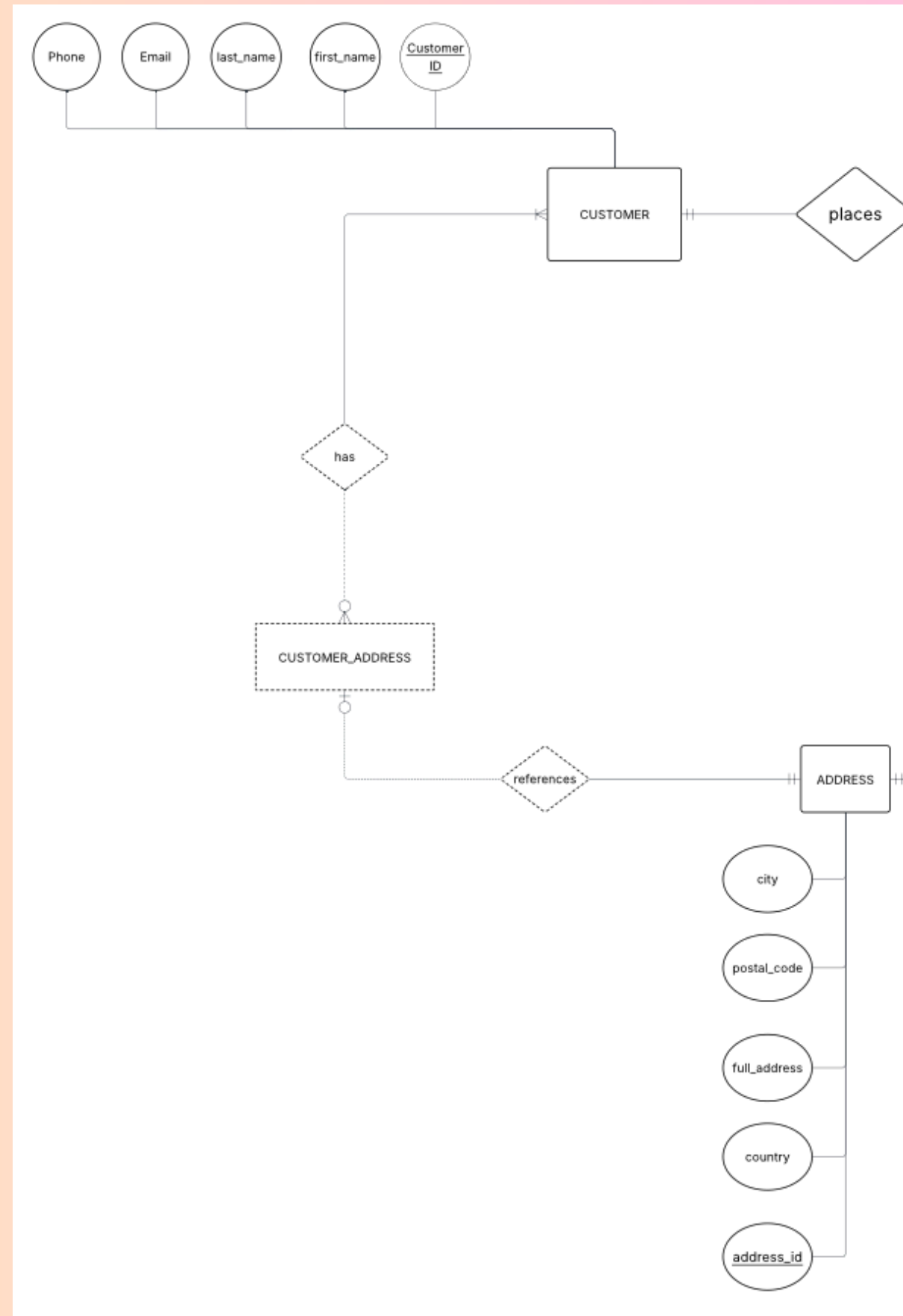
Weak Entity: Dependent entities that resolve relationships and cannot exist independently.

- Examples in our schema: CUSTOMER_ADDRESS, ORDER_MENU_ITEM

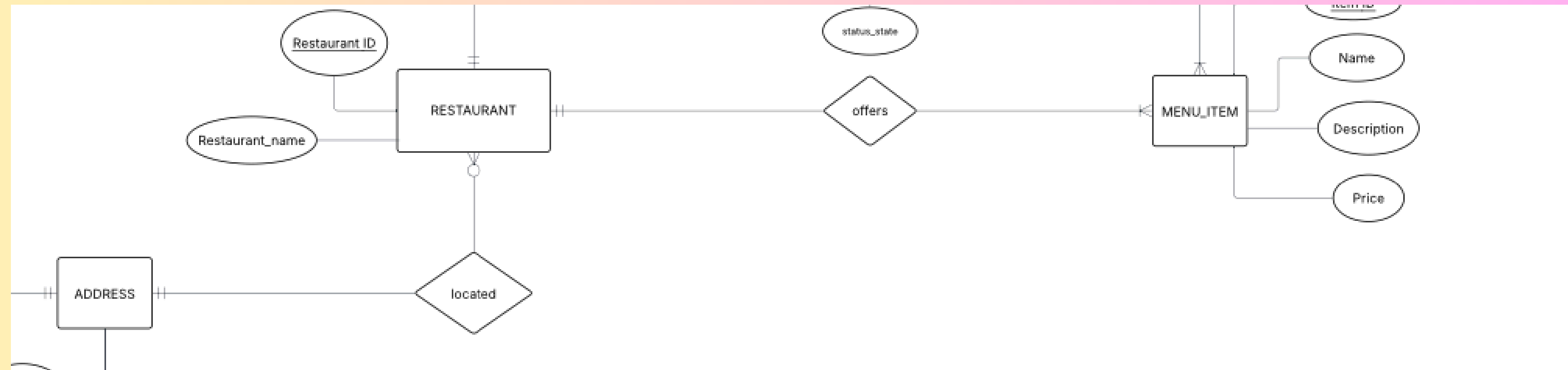
ER Diagram- Food Delivery System



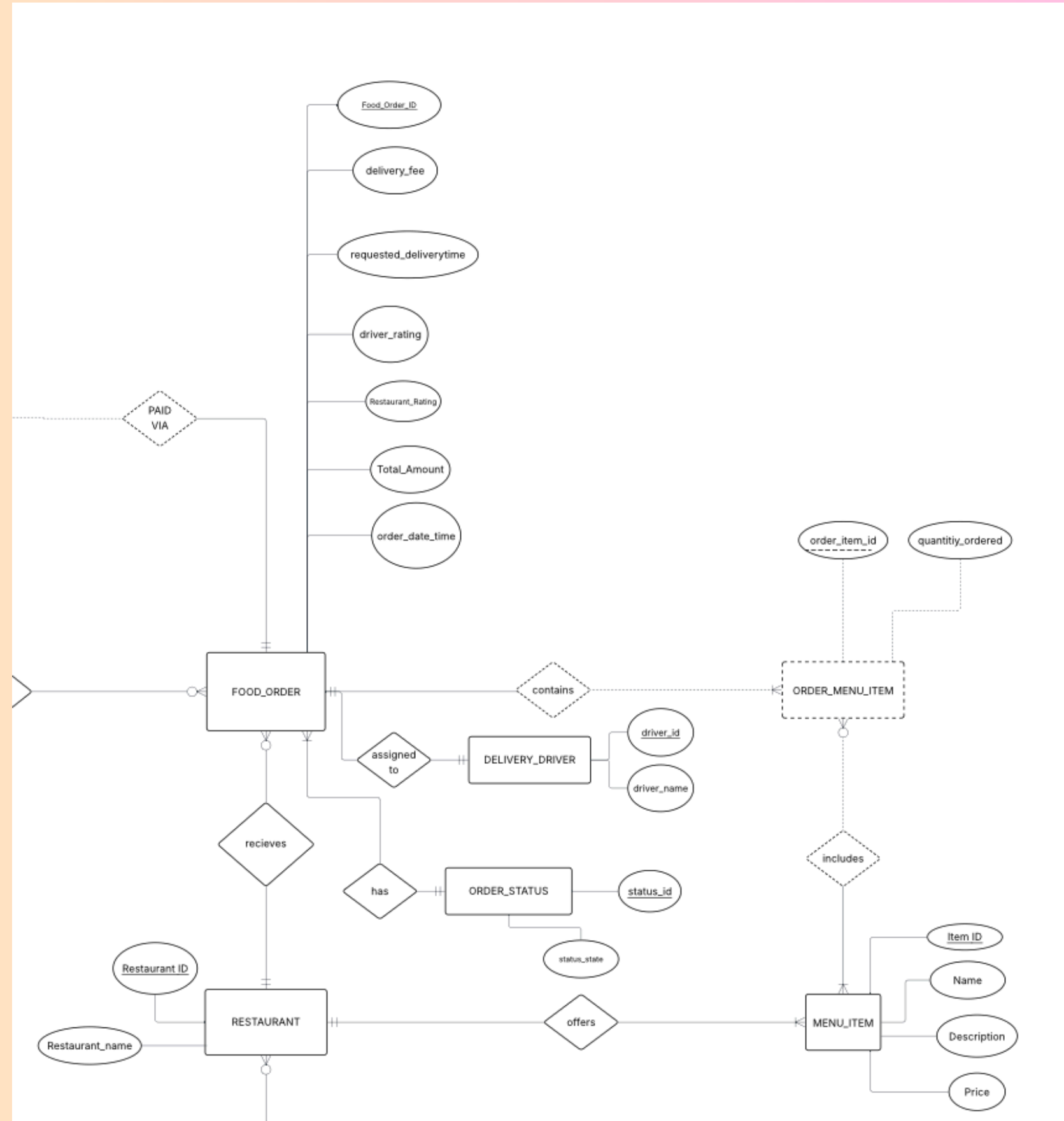
USER PROFILE & ADDRESS MANAGEMENT



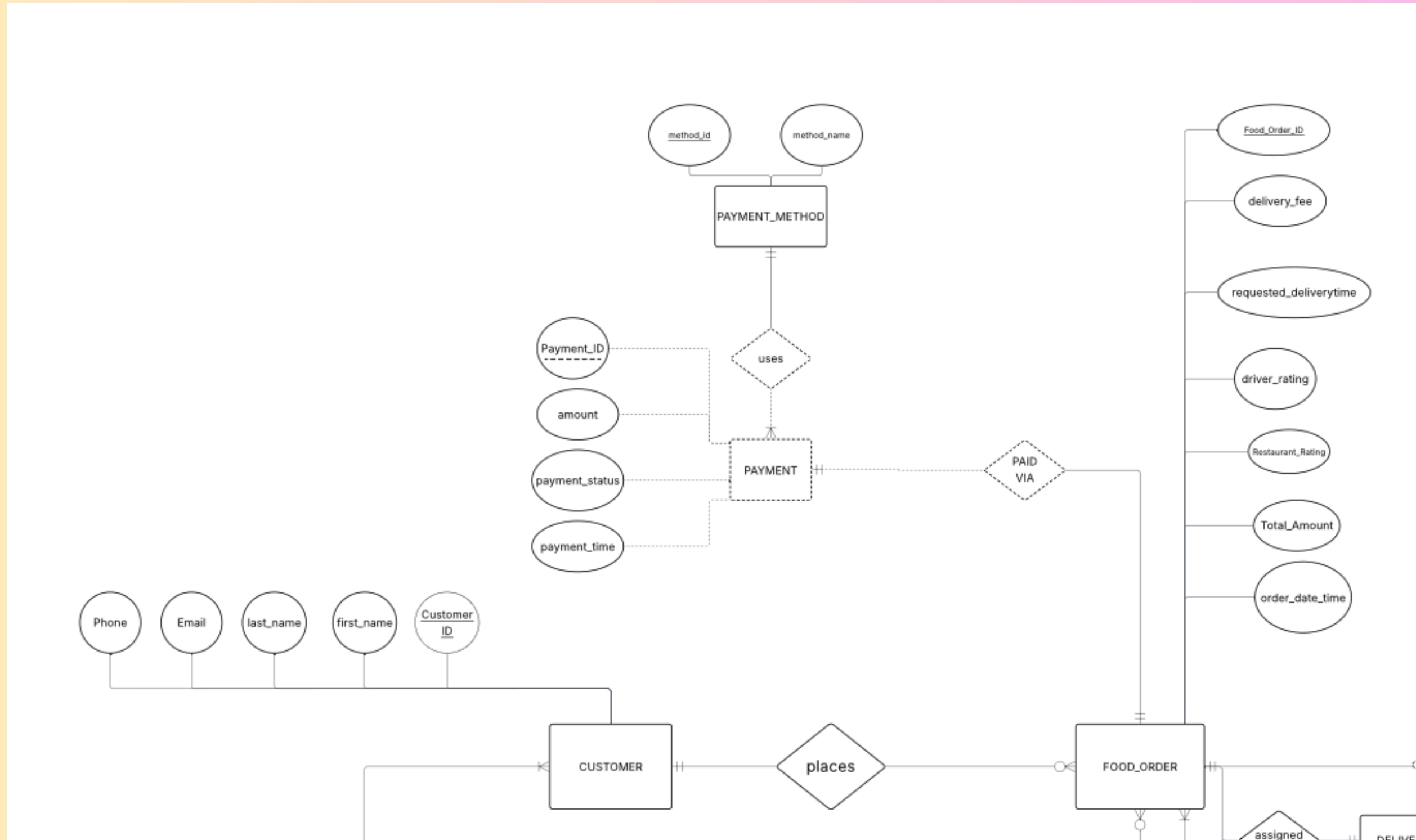
RESTAURANT & MENU CATALOGUE



ORDER & DELIVERY WORKFLOW



PAYMENT & TRANSACTION DETAILS



RELATIONAL SCHEMA

